

Review I.



What have we learned so far.

1. division algorithm
2. Euclidean algorithm
3. Bezout's theorem.
4. congruence.
5. prime & irreducibles.

Since $85 - 6 \cdot 14 = 1$

$$85 \cdot 3 + 14 \cdot (-18) = 3$$

i.e. $x=3$, $y=-18$ is a particular solution

Suppose x' , y' is another pair of solution

So $85x' + 14y' = 3$

$$85 \cdot 3 + 14 \cdot (-18) = 3.$$

Then. $85 \cdot (x'-3) = 14 \cdot (-y'-18)$

By Euclidean lemma,

$$14 \mid x'-3 \quad \text{and} \quad 85 \mid -y'-18$$

Write $x' = 3 + 14k$ $k \in \mathbb{Z}$
 $y' = -18 - 85p$ $p \in \mathbb{Z}.$

eg. Solve $8x \equiv 1 \pmod{6}$

If x_0 is a solution, then.

$$8x_0 - 1 \equiv 0 \pmod{6}.$$

$$\text{i.e. } \exists y_0 \in \mathbb{Z} \text{ s.t. } 8x_0 - 1 = 6y_0$$

$$\text{i.e. } 8x_0 - 6y_0 = 1$$

But $2 \mid 8x_0 - 6y_0$ but $2 \nmid 1$.

So no solutions

eg. Solve. $8x \equiv 4 \pmod{6}$.

Similarly, if x_0 is a particular solution.

then $\exists y_0 \in \mathbb{Z}$ s.t. $8x_0 - 6y_0 = 4$

Since $(8, 6) = 2 = 8 - 6$.

We know $8 \cdot 2 - 6 \cdot 2 = 4$

and x_0 is a particular solution.

If x' is another solution.

$8x' \equiv 4 \pmod{6}$, $8 \cdot 2 \equiv 4 \pmod{6}$

implies $8(x' - 2) \equiv 0 \pmod{6}$

$6 \mid 8(x' - 2) \Rightarrow 3 \mid x' - 2$.

So all solutions are $2 + 3k$, $k \in \mathbb{Z}$.

eg. Solve $3x + 6y + 20z = 40$

for $x, y, z \in \mathbb{N}$.

Take mod 3 on both sides of the equation

$$2z \equiv 1 \pmod{3}$$

$$z \equiv 2 \pmod{3}$$

Write $z = 2 + 3k$. for $k \geq 0$

but $k \leq 0$ for otherwise $2z > 40$.

It is clear there is no solutions in $\mathbb{N}$.

Note we get

0, 3, ~~6, 9, 12, 15, 18, 21~~, ~~24~~ 39

1, 4, 7, 10, 13, 16, 19, 40

2, 5, 8, 11, 14, 17, 20, ~~23~~ 41.

How to land in $[1]$?

$$20 \times 2 = 40 \equiv 1 \pmod{3}$$

This is the smallest $\equiv 1 \pmod{3}$

in the set $\{3x + 6y + 20z : x, y, z \in \mathbb{N}\}$.

So $N = 39$.

solution to
$$\begin{cases} x \equiv 4 \pmod{5} & \textcircled{1} \\ x \equiv 2 \pmod{6} & \textcircled{2} \end{cases}$$

Now write $x = 5k + 4$ for some $k \in \mathbb{Z}$.

Plug in to $\textcircled{2}$, $5k + 4 \equiv 2 \pmod{6}$

$$-k \equiv -2 \pmod{6} \quad \text{iff} \quad k \equiv 2 \pmod{6}$$

So write $k = 6p + 2$ for $p \in \mathbb{Z}$

we get $x = 5(6p + 2) + 4$

$$= 30p + 14 \quad \text{for } p \in \mathbb{Z}$$

It is easy to verify for each $p \in \mathbb{Z}$.

$x = 30p + 14$ is a solution. So it's all sols.